

Assignment 2: Policy Gradient

Andrew ID: Write your Andrew ID here.

Collaborators: Write the Andrew IDs of your collaborators here (if any).

NOTE: Please do NOT change the sizes of the answer blocks or plots.

5 Small-Scale Experiments

5.1 Experiment 1 (Cartpole) – [5 points total]

5.1.1 Configurations

Q5.1.1

```
python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
-dsa --exp_name q1_sb_no_rtg_dsa

python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
-rtg -dsa --exp_name q1_sb_rtg_dsa

python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
-rtg --exp_name q1_sb_rtg_na

python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 5000 \
-dsa --exp_name q1_lb_no_rtg_dsa

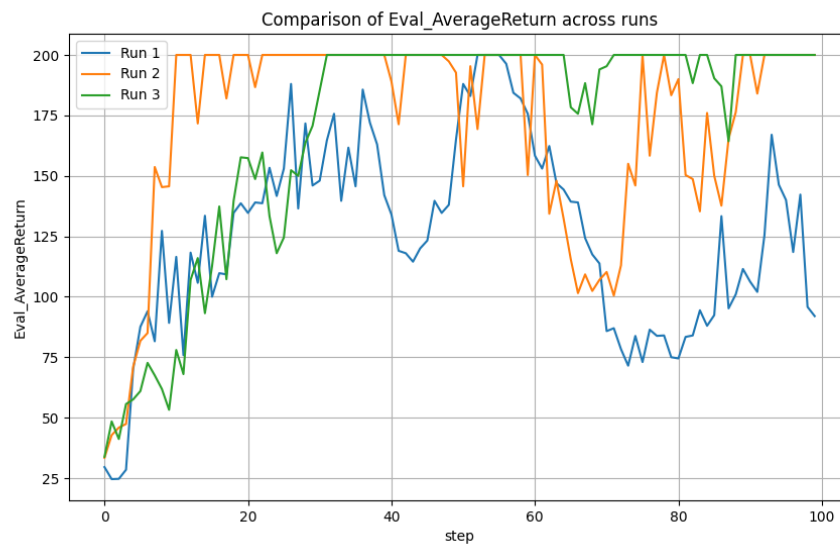
python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 5000 \
-rtg -dsa --exp_name q1_lb_rtg_dsa

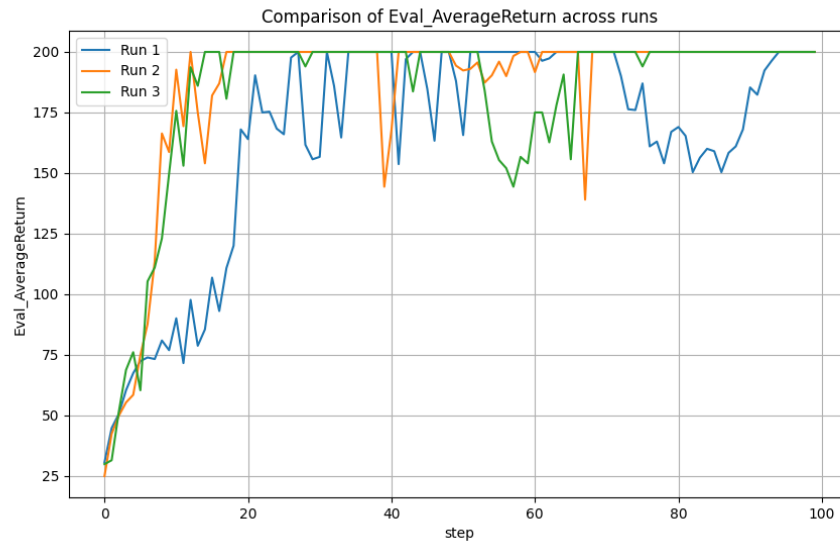
python rob831/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 5000 \
-rtg --exp_name q1_lb_rtg_na
```

5.1.2 Plots

5.1.2.1 Small batch – [1 points]

Q5.1.2.1



5.1.2.2 Large batch – [1 points]**Q5.1.2.2****5.1.3 Analysis****5.1.3.1 Value estimator – [1 points]****Q5.1.3.1**

The reward-to-go estimator generally performs better than the full trajectory and the improvement is significant.

5.1.3.2 Advantage standardization – [1 points]**Q5.1.3.2**

Advantage standardization helps stabilize training and leads to better performance and the improvement is significant.

5.1.3.3 Batch size – [1 points]**Q5.1.3.3**

Larger batch size leads to more stable training and better performance but the improvement is not very significant.

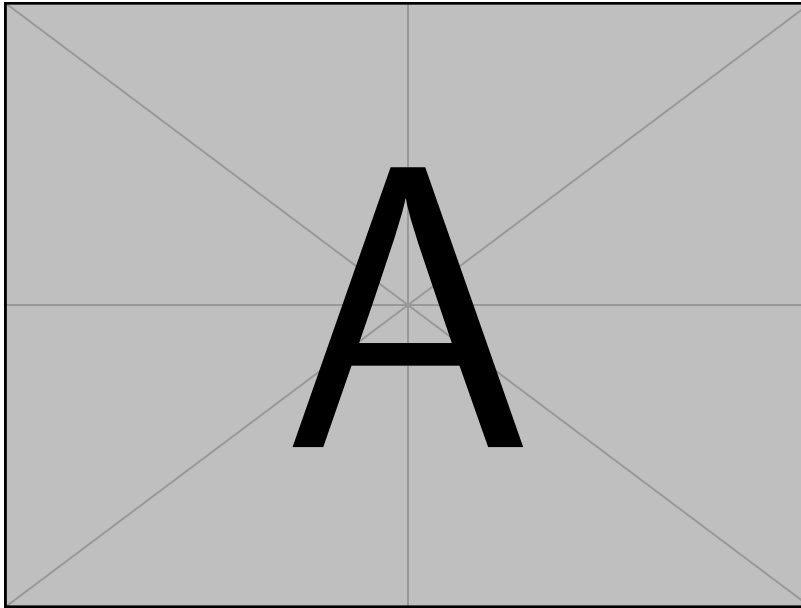
5.2 Experiment 2 (InvertedPendulum) – [4 points total]**5.2.1 Configurations – [1.5 points]****Q5.2.1**

```
python rob831/scripts/run_hw2.py --env_name InvertedPendulum-v4 \
--ep_len 1000 --discount 0.9 -n 100 -l 2 -s 64 -b <b*> -lr <r*> -rtg \
--exp_name q2_b<b*>_r<r*>
```

5.2.2 smallest b^* and largest r^* (same run) – [1.5 points]**Q5.2.2**

5.2.3 Plot – [1 points]

Q5.2.3



7 More Complex Experiments

7.1 Experiment 3 (LunarLander) – [1 points total]

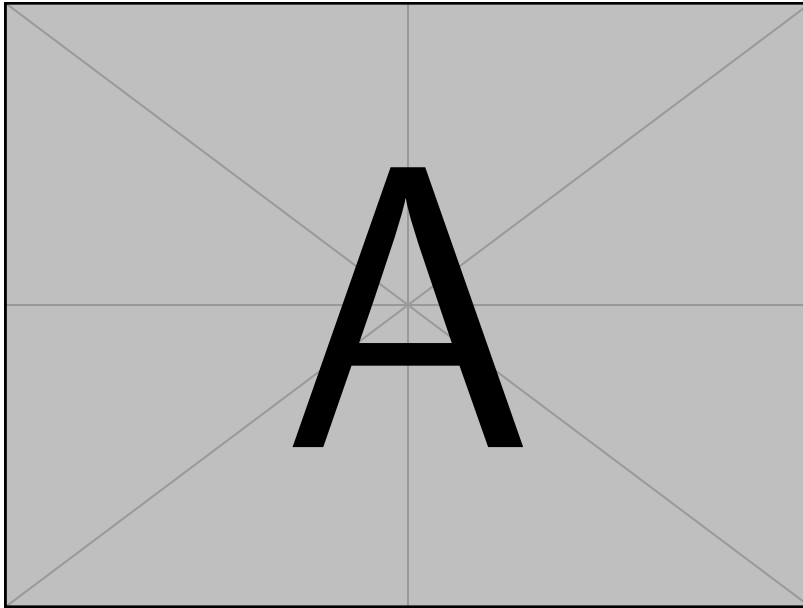
7.1.1 Configurations

Q7.1.1

```
python rob831/scripts/run_hw2.py \  
  --env_name LunarLanderContinuous-v4 --ep_len 1000 \  
  --discount 0.99 -n 100 -l 2 -s 64 -b 10000 -lr 0.005 \  
  --reward_to_go --nn_baseline --exp_name q3_b10000_r0.005
```

7.1.2 Plot – [1 points]

Q7.1.2

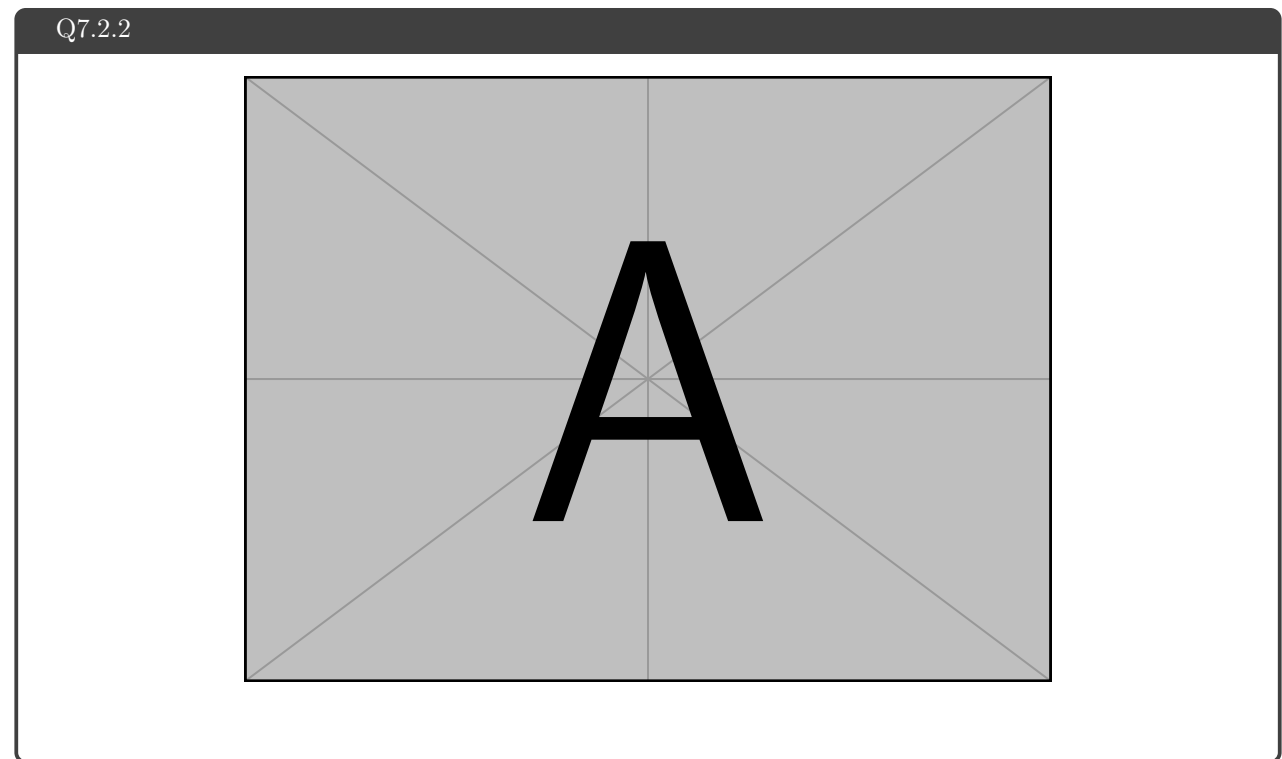


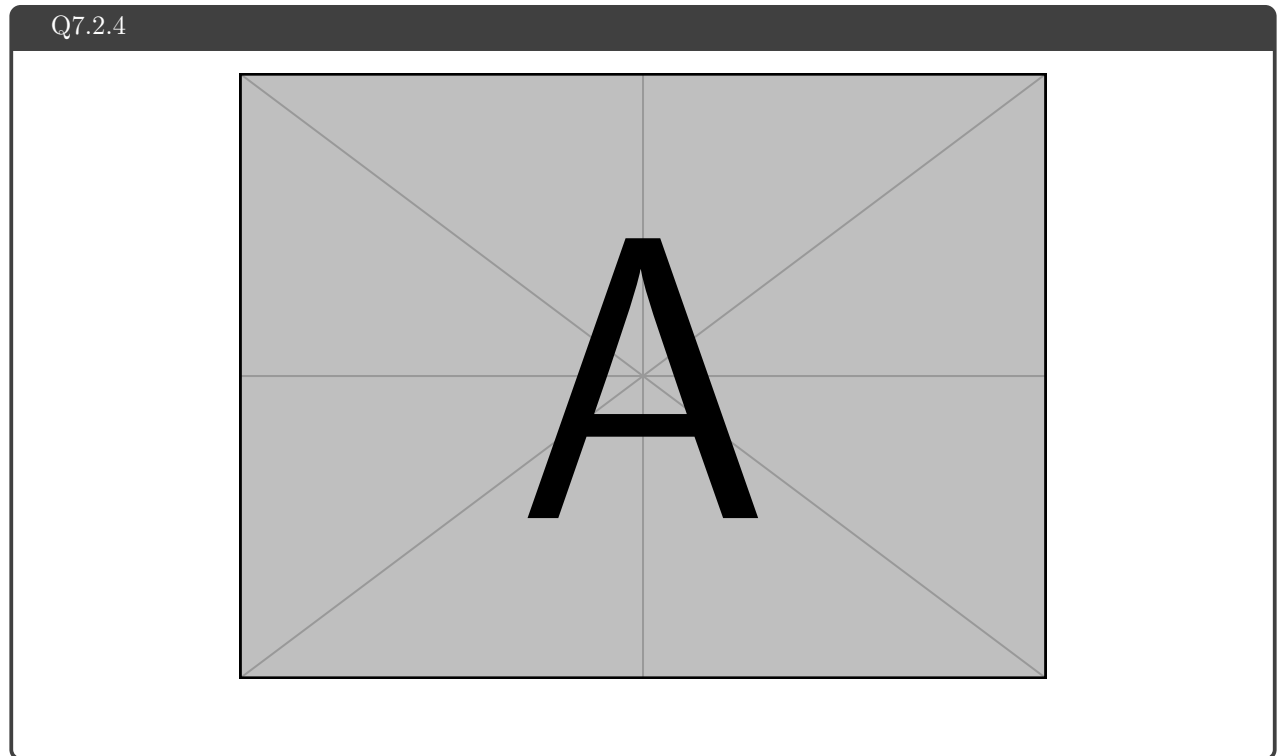
7.2 Experiment 4 (HalfCheetah) – [1 points]

7.2.1 Configurations

Q7.2.1

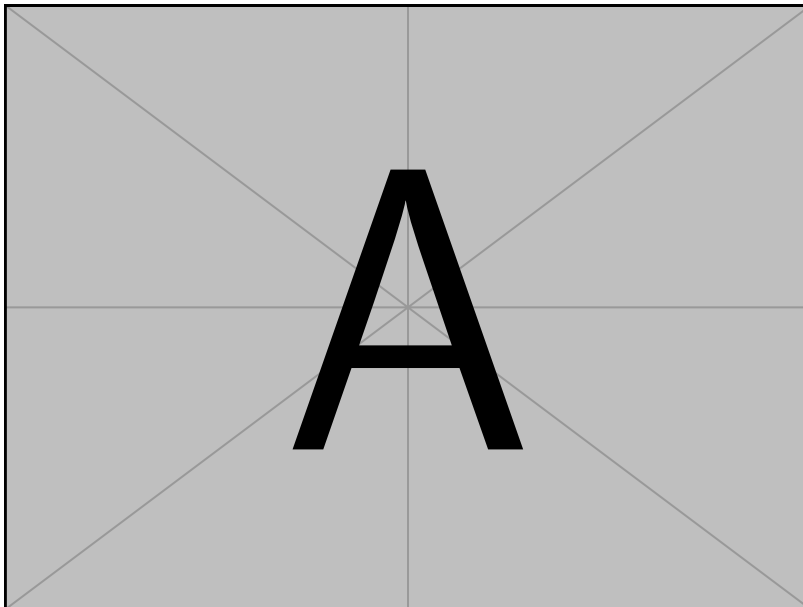
```
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
  --discount 0.95 -n 100 -l 2 -s 32 -b 10000 -lr 0.02 \  
  --exp_name q4_search_b10000_lr0.02  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
  --discount 0.95 -n 100 -l 2 -s 32 -b 10000 -lr 0.02 -rtg \  
  --exp_name q4_search_b10000_lr0.02_rtg  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
  --discount 0.95 -n 100 -l 2 -s 32 -b 10000 -lr 0.02 --nn_baseline \  
  --exp_name q4_search_b10000_lr0.02_nnbaseline  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
  --discount 0.95 -n 100 -l 2 -s 32 -b 10000 -lr 0.02 -rtg --nn_baseline \  
  --exp_name q4_search_b10000_lr0.02_rtg_nnbaseline
```

7.2.2 Plot – [1 points]**7.2.3 (bonus) Optimal b^* and r^* – [0.5 points]**

7.2.4 (bonus) Plot – [0.5 points]**7.2.5 (bonus) Describe how b^* and r^* affect task performance – [0.5 points]**

7.2.6 (bonus) Configurations with optimal b^* and r^* – [0.5 points]**Q7.2.6**

```
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
--discount 0.95 -n 100 -l 2 -s 32 -b <b*> -lr <r*> \  
--exp_name q4_b<b*>_r<r*>  
  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
--discount 0.95 -n 100 -l 2 -s 32 -b <b*> -lr <r*> -rtg \  
--exp_name q4_b<b*>_r<r*>_rtg  
  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
--discount 0.95 -n 100 -l 2 -s 32 -b <b*> -lr <r*> --nn_baseline \  
--exp_name q4_b<b*>_r<r*>_nnbaseline  
  
python rob831/scripts/run_hw2.py --env_name HalfCheetah-v4 --ep_len 150 \  
--discount 0.95 -n 100 -l 2 -s 32 -b <b*> -lr <r*> -rtg --nn_baseline \  
--exp_name q4_b<b*>_r<r*>_rtg_nnbaseline
```

7.2.7 (bonus) Plot for four runs with optimal b^* and r^* – [0.5 points]**Q7.2.7****8 Implementing Generalized Advantage Estimation**

8.1 Experiment 5 (Hopper) – [4 points]

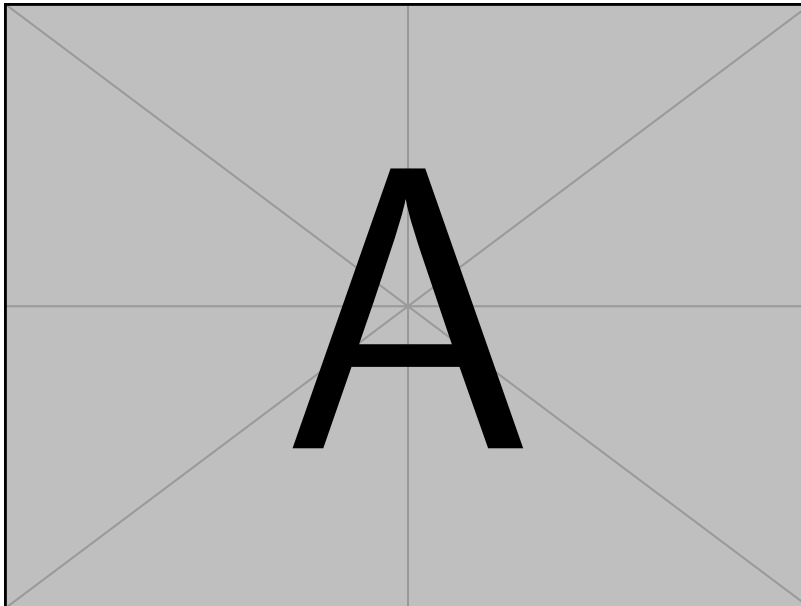
8.1.1 Configurations

Q8.1.1

```
#  $\lambda \in [0, 0.95, 0.99, 1]$ 
python rob831/scripts/run_hw2.py \
  --env_name Hopper-v4 --ep_len 1000
  --discount 0.99 -n 300 -l 2 -s 32 -b 2000 -lr 0.001 \
  --reward_to_go --nn_baseline --action_noise_std 0.5 --gae_lambda < $\lambda$ > \
  --exp_name q5_b2000_r0.001_lambda< $\lambda$ >
```

8.1.2 Plot – [2 points]

Q8.1.2



8.1.3 Describe how λ affects task performance – [2 points]

Q8.1.3

9 More Bonus!

9.1 Parallelization – [1.5 points]

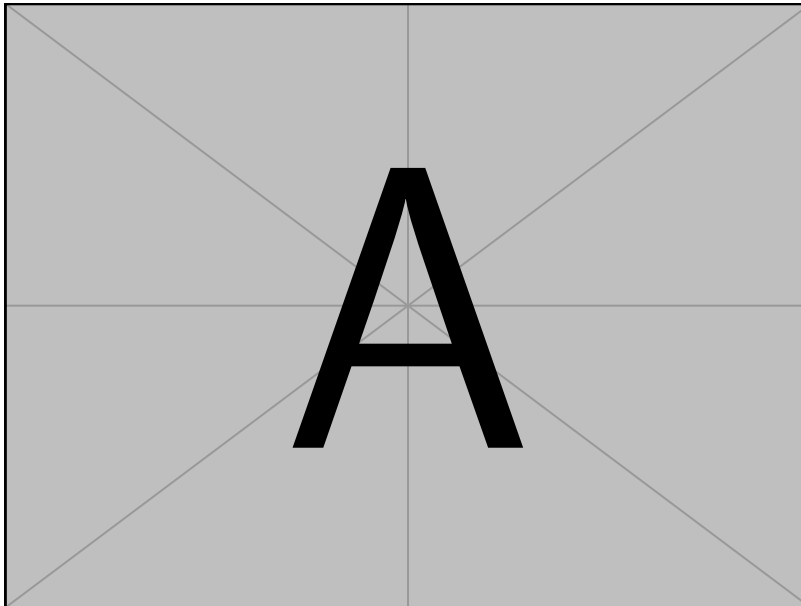
Q9.1

Difference in training time:

```
python rob831/scripts/run_hw2.py \
```

9.2 Multiple gradient steps – [1 points]

Q9.1



```
python rob831/scripts/run_hw2.py \
```